The Arithmetic of Suspicion

A statistical analysis of how the Thirlwall Inquiry's recommendations on monitoring and suspicion could interact

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In brief

Recommendation 6 of the Thirlwall Inquiry requires trusts to monitor deaths; Recommendation 9 requires them to act on any good-faith suspicion that a member of staff has deliberately harmed a patient. Nothing governs the step between them: *how a trust moves from an outcome anomaly to a targeted investigation of a specific person*. The expert review proposed in Recommendation 10 begins only once a person is already suspected. This article measures what that gap costs, under explicit scenarios, and sets out the statistical safeguard it needs. It is not a forecast of how trusts will implement the recommendations.

Why a simulation?

Because deliberate harm is so rare, the historical record cannot show how often this step goes wrong. Where a case rests on direct evidence, such as a confirmed poisoning or a witness, the cause is known. But the record holds only cases that reached a court or a formal inquiry, and it scores them by their outcome: a conviction counts as a confirmed offender, so a wrongful conviction built on a misread pattern enters the record as a success. The record also omits the chance alarms that led nowhere, the nurses suspected and quietly cleared, and the offenders never caught. The error rate of the move from alarm to suspect cannot be measured from a record that counts its own mistakes as correct. A simulation can measure it, because it builds units where the truth is known: which deaths are chance, and whether an offender is present. Thirlwall Inquiry Recommendation 6 requires monitoring but does not say what counts as a concerning trend, so this model tests six candidate rules: four simple rules a trust might adopt, from an annual count to a rolling monthly check, and, as a best case, an idealised risk-adjusted chart of the kind the report itself describes (Table 1). The simulation then follows the sequence the recommendations leave open, in three steps:

1. how often chance alone triggers each rule;
2. how often a rota search after a chance alarm makes one nurse look significant;
3. how those false signals compare with genuine ones, ending with the chance that a flagged nurse is in fact the offender.

What the results measure is the cost of that unguarded step, and so what a safeguard must protect against.

The scale of the problem

Consider a neonatal unit with 50 nurses, roughly the average in England and Wales. Deaths fluctuate by chance, so a rule for detecting unusually high mortality will sometimes be triggered when no offender is present. Suppose chance triggers it once every ten years. That is the rate the simulation sets for the idealised chart in Table 1 (E10), and lies within the range the simpler rules show in simulation for a local unit: once every three years to once every 80 years. The UK conviction record suggests that a nurse who deliberately harms patients is at work for roughly one year in every 5 million nurse-years, or one year in 100,000 for a 50-nurse unit. Over that period, the rule would be triggered about 10,000 times with no offender present. Even if it caught every offender-year, an alarm would be about 10,000 times more likely to occur without an offender than with one. No rota has yet been examined.

The model behind this article refines that rough sum. It simulates the 175 neonatal units in England and Wales under each of the six alarm rules, which differ in how often they are triggered, and it allows for the fact that an offender does not always raise the alarm. It counts two kinds of year: a *falsely flagged unit-year*, in which a unit with no offender is flagged, and a *detected offender-year*, in which an offender is at work and the unit's alarm is triggered. "Detected" means only that the alarm was triggered, not that anyone was identified. The result is **about 11,000 to 49,000 falsely flagged unit-years for every detected offender-year**, depending on the rule. Taking the single neonatal case at face value implies an offending rate ten times higher, and the figure is still about 1,100 to 4,900.

The obvious remedy is a stricter alarm, one that needs stronger evidence before it is triggered. That does cut false alarms, but it also lets most real cases pass unnoticed (Figure 2). At the strictest setting tested, even at the higher rate, there are still about 400 false alarms for every detection, and across all 175 units a detection occurs **about once every 23,000 years**.

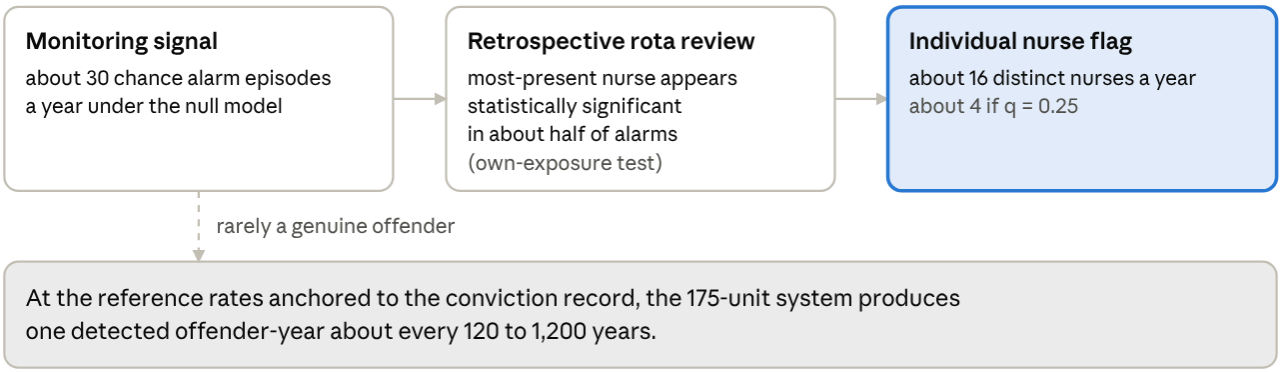
From alarm to suspect

A second source of error arises if a mortality signal is followed by a retrospective search of the rota. Someone will always be the nurse most often present, and under a naive significance test, that nurse appears statistically significant in about half of chance alarms. Yet when a real offender is present, the same rota exercise can fail to identify the offender.

One concrete example highlights this. Under one of those rules, a rolling monthly check (rule C in Table 1), chance alone produces about 30 alarm episodes a year. If every alarm prompts a rota search, about 16 distinct nurses a year acquire a nominally significant rota record (Figure 1).

Figure 1. Chance mortality alarms can flag about 16 individual nurses a year

Rolling monthly monitoring (Rule C), England and Wales, if every alarm is followed by a rota review (q = 1)



Simulation details: 175 units · rule C · base rates 0.1 and 1 in 10,000 unit-years, risk in proportion to staff

The alternative to weak monitoring is not broader reporting but stronger investigation. A mortality signal should bring skilled investigators into the unit to examine staffing, workload, infection, equipment, clinical practice and case mix. Only evidence uncovered by that investigation should move the question from “what happened here?” to “did somebody do this?”

About this article

This article works through the arithmetic behind this distinction: how often monitoring alarms by chance, what happens when those alarms are matched retrospectively to staff, how much evidential weight the resulting pattern actually carries, and where a statistical safeguard needs to sit if surveillance is not to generate suspicion more often than it detects wrongdoing. Its figures come from a single, reproducible run of the model. An accompanying interactive explorer reproduces them exactly and lets readers change the assumptions behind each table, from unit mortality and staffing to alarm thresholds and offender prevalence, and follow the consequences through to the nurses flagged.

How the recommendations fit together

Read together, three of the recommendations leave a potentially important gap. The report provides an independent expert safeguard once concerns about an individual have emerged, but it does not specify a statistical safeguard governing the prior step: how an

abnormal outcome pattern becomes suspicion of a person. That is the critical inferential transition. A mortality signal may be sufficient reason to investigate a unit; it is not, without further evidence, sufficient reason to focus that investigation on an individual.

The recommendations create a clear sequence, but the statistically important step sits between them.

1. **Monitoring.** Recommendation 6 requires Board-level monitoring of deaths of children and babies, with a predetermined route for escalating concerning data trends. It does not prescribe what should count as a concerning trend. The report itself recognises the limitation: as Sir David Spiegelhalter told the Inquiry, a monitoring system "cannot say why something has happened".
2. **Suspicion.** Recommendation 9 concerns what happens once a healthcare professional is suspected of deliberate harm. The threshold is deliberately low: a good-faith concern must be acted on unless it is obviously irrational or malicious. The recommendation therefore governs the response to suspicion; it does not establish how suspicion should be formed.
3. **Action.** Under Recommendation 9, once the concern has attached to an individual, protective action may follow before its cause has been established. The report treats removal from clinical duties as a neutral measure. Whatever its legal or managerial status, however, it is a consequential step for the person concerned and for the unit in which they work.
4. **Expert review.** Recommendation 10 asks DHSC to consider a panel capable of carrying out rapid technical investigations where concerns involve an individual and possible harm. That safeguard operates downstream: it begins after the investigation has already moved from an abnormal outcome to a particular person. The examples of expertise given do not specifically include a statistician, although other experts could of course be added.

The statistical problem lies in the transition between the first two stages. **How does a signal about outcomes become a suspicion about a person?** If a mortality alert leads to a retrospective search of the rota, the process has crossed from population-level surveillance to individual attribution. That step has its own error rate, and it cannot be inferred from the mortality chart itself.

The simulation therefore asks three questions: how often chance alone produces an alarm; how often a retrospective rota search then produces an apparently significant nurse; and how much evidential weight that combination carries once the rarity of deliberate harm is taken into account. How often managers actually make that transition is behavioural rather than statistical, so the model does not assume an answer. It leaves that probability, q, explicit.

The analytical map

The analysis proceeds through the sequence from mortality monitoring to individual attribution. Each section isolates one stage of that sequence and the statistical questions it raises.

Section	Analytical question	Main finding
Step one: over the threshold	How often will ordinary variation cross a monitoring threshold?	Depending on the rule, chance alone produces about 3–34 alarms a year across England and Wales. At the two benchmark settings tested, even the idealised, perfectly risk-adjusted monitor produces about 4–18.
Step two: the rota review	What happens when alarm-associated deaths are matched retrospectively to staff attendance?	Tested as if named in advance, the most-present nurse appears statistically significant in about half of chance alarms; allowing for selection from the whole rota reduces this to about 2% under the simulated null.
Step three: from signal to evidence	How much evidential weight does an alarm carry once prevalence and detection sensitivity are taken into account?	At the two reference rates anchored to the conviction record, the model produces roughly 1,000–50,000 falsely flagged unit-years for each detected offender-year. Stricter thresholds reduce the ratio only at the cost of sharply lower detection sensitivity, while retrospective rota analysis can also fail to identify a genuine offender.

Section	Analytical question	Main finding
Costing the consequences	What follows if chance signals lead to person-focused action?	Staff may be removed from units already operating under staffing pressure, while camera schemes operate at substantial scale for a very low expected detection yield.
Implications for practice	What should a mortality signal trigger?	An independent review of care, not a search for a person: thresholds set nationally in advance, staffing, case mix and clinical systems examined first, and any later analysis of attendance covering the complete rota, blind.
Synthesising the signals	What follows when the three statistical stages are considered together?	Monitoring can identify an abnormal outcome pattern, but it cannot establish deliberate harm or identify its cause. The principal statistical vulnerability lies in the transition from an outcome signal to concern about a particular person.
Assumptions and limits	Which parts of the result depend on modelling assumptions?	The analysis does not predict managerial behaviour: how often managers move from an alarm to a named person is represented by an explicit parameter, q. Rotas are simplified and prevalence is examined through scenarios because the historical record cannot identify a precise base rate.

Step one: over the threshold

Depending on the rule, chance variation alone produces between 3 and 34 alarms a year across England and Wales, with no change in the quality of care.

Small counts are inherently volatile. A local unit expecting four deaths a year will record eight in some years with no change in care. At these numbers, distinguishing ordinary variation from a genuine shift entails a substantial trade-off between false alarms and sensitivity.

The model. It simulates 175 neonatal units in England and Wales: roughly the UK's 191 neonatal services (RCPCH, 2019), less Scotland's 15 (Scottish Perinatal Network) and Northern Ireland's handful. They are of three types:

- 45 intensive care units, averaging 20 deaths a year
- 85 local units, averaging 4
- 45 special care units, averaging 1

That gives about 1,300 deaths a year on units, roughly three-quarters of the approximately 1,770 neonatal deaths in England and Wales in 2023, a rate of 3.0 per 1,000 live births (ONS); the remainder occur outside neonatal units. Each unit's case mix varies from year to year (drawn from a gamma distribution with a 25% coefficient of variation), and rules A–D compare a unit with its own average over the three years before the window being checked, with a floor of half a death a year so that a unit with no recent deaths is not alarmed by a single one. Each unit type was simulated 2,000 times over ten years of monitoring.

The report does not say what counts as a "concerning trend", so I tested four simple rules a trust might adopt, together with an idealised benchmark (Table 1). The 2- and 3-standard-deviation lines are shorthand: each is a one-sided cut-off at the matching percentile (97.7% or 99.87%) of a Poisson count whose mean is the unit's three-year average.

Table 1. Simulated chance alarms per year across England and Wales, by monitoring rule

Rule	What triggers it	Unit-years with an alarm, per year	Distinct alarm episodes, per year
A	Year's deaths above the usual 2-standard-deviation line, checked once a year	11	11
B	The stricter 3-standard-deviation line, checked once a year	3	3
C	2-standard-deviation line on a rolling 12 months, checked every month	32	30

Rule	What triggers it	Unit-years with an alarm, per year	Distinct alarm episodes, per year
D	A deliberately simple heuristic: rolling 12 months shows deaths have doubled (at least 4), checked monthly	34	34
E10	Idealised benchmark: a perfectly risk-adjusted CUSUM chart tuned to a doubling of deaths, set to 1 false alarm per 10 unit-years	17	18
E50	The same chart set to 1 false alarm per 50 unit-years	4	4

Rule D falls most heavily on local units, alarming in about one unit-year in three. Rule C shows the cost of monthly checking: each additional check is a further opportunity for random variation to cross the threshold. Counting distinct episodes rather than unit-years makes little difference, because runs of alarms seldom span two years.

Rules E10 and E50 address the objection that real monitoring would outperform rules A to D. The report itself describes risk-adjusted sequential charts of the kind PICANet uses for paediatric intensive care (Chapter 35). Rule E is an idealised version: it knows each month's true expected deaths, so changes in case mix cannot trigger it; only random variation around that expectation can. How often that happens is not a finding but a setting, fixed in advance, by simulating units with no offender, at two operating points. It can be set stricter still, at the cost of lower detection; step three shows what that trade-off buys. It is tuned to a doubling of deaths, so it is not the optimal detector for every alternative in every kind of unit, but it is a demanding benchmark.

Removing the assumed year-to-year variation reduces the alarm rate but does not eliminate chance alarms: with perfect knowledge of each unit's true death rate, rules C and D still produce about 13 and 16 a year.

The Shipman Inquiry met the same problem. When Imperial College researchers ran a monitoring chart over 1,009 GPs, 33 crossed the threshold; Shipman was one (Aylin et al., Lancet 2003; reported by politics.co.uk). The researchers judged that the others were almost certainly statistical or medical false alarms (Plus magazine). A later study found that such charts only reliably detected a doctor causing 10 excess deaths a year, while raising false alarms for more than 15% of practices (Guthrie et al., BJGP 2008). The chart did flag Shipman, whose excess deaths, around ten a year over more than two decades, were large by any standard; it could not distinguish him from the 32 others, and only investigation did. On its own, a chart indicates where to look, not at whom.

Step two: the rota review

A mortality alarm identifies a set of events, not an individual. Once those events are matched retrospectively against staff attendance, however, the analysis acquires a second source of false positives. In the simulation, a nurse appears statistically significant in about half of chance alarms if the most-present nurse is tested as though that nurse had been identified in advance.

The problem is selection. In a rota of 40 or 100 nurses, one nurse must have the greatest overlap with the deaths even when deaths and staff attendance are entirely unrelated. The relevant question is therefore not whether **that nurse's** overlap looks unusual in isolation, but whether the **largest overlap anywhere on the rota** is unusual. Treating the selected nurse as if specified beforehand understates the probability of obtaining an apparently striking result by chance.

The model. Every chance alarm episode from step one, with consecutive alarmed checks counted as one, is followed by a review of the deaths in the window that generated it, provided that window holds at least two deaths, as almost all do. Staff numbers vary with unit size: 100 nurses in an intensive care unit, 40 in a local unit and 20 in a special care unit. Half work a full-time pattern of three of the week's fourteen 12-hour shifts, giving an attendance probability of about 21%; a third work part time, at about 13%; and the remainder work more frequently, at about 27%. Deaths are assigned independently of staff attendance: no nurse causes any of them.

I evaluated the most-present nurse under three progressively better-specified tests:

1. **Average-exposure test.** The nurse's attendance is compared with the average nurse's share of shifts. This ignores variation in individual exposure and is essentially the inference invited by a bare statement that someone was present for "X of Y" events.
2. **Own-exposure test.** The nurse's attendance is compared with that nurse's own share of shifts. This corrects for working pattern, but still treats the nurse as if named before the rota was examined.
3. **Maximum-adjusted test.** The comparison is made against the most extreme attendance that would be expected anywhere on the rota, given every nurse's actual share of shifts. Under the simulated null, where attendance is independent across nurses at their own rates, this probability can be calculated exactly. A real rota, with fixed staffing, teams and shift structure, would require a null model built from the rota itself.

Table 2. Rota-review results for Rule C chance alarms, by unit type

Unit type (rule C alarms)	Deaths reviewed (median)	Top nurse attended (median)	Looks significant: average-exposure test	Own-exposure test	Maximum-adjusted test (under this model)
Intensive care	28	12	100%	69%	3.1%
Local	8	4	68%	51%	1.8%
Special care	3	2	26%	26%	1.4%
All reviewed episodes across the 175 units	—	—	76%	55%	2.3%

The maximum-adjusted test behaves as it should when the deaths are random: it flags only about 2%, below the nominal 5% because attendance counts are whole numbers. The own-exposure test finds a "significant" nurse in more than half of all alarms, because no one was named in advance. The nurse was picked *because* that nurse's record was the highest of 40 or 100. This is a standard post-selection error, sometimes called the Texas sharpshooter fallacy: the most extreme observation is selected first and then tested as though it had been specified in advance.

The average-exposure test does worse still, because the nurse who tops the rota is usually one who works more shifts than average. It is a second, separate error, but a common one: a rota chart with no denominator commits it by default. The other rules give much the same picture: across all six, the own-exposure test flags a nurse in 49% to 60% of alarms, and the maximum-adjusted test in about 3% or fewer.

The model also omits several forms of analytic flexibility that can further inflate apparent associations. The Royal Statistical Society report on these cases identifies three (Green, Gill et al. 2022):

- **Broadened event definitions.** Collapses, desaturations and "unexpected deteriorations" are added to the deaths, multiplying the events matched against the rota.
- **Flexible windows.** The period under review is chosen after the cluster is seen.
- **Selective counting.** Once an individual has been named, the events that nurse missed tend to be explained away and those attended retained.

Step three: from signal to evidence

Step one showed how often chance triggers an alarm; step two, how readily a rota search attaches a name to it. Step three sets those false signals against genuine ones: first for units, comparing falsely flagged unit-years with detected offender-years, and then, at the end of the section, for the nurses named.

At the two reference rates anchored to the UK conviction record, 0.1 and 1 offender-year per 10,000 unit-years, about 1,000 to 50,000 unit-years are falsely flagged for each detected offender-year. Even at 30 per 10,000, a deliberately high-prevalence stress test, the ratios span about 35 to 165 across these monitoring rules.

This is a base-rate problem. Deliberate harm is so rare that even a monitor with a low false-alarm rate will encounter many more innocent fluctuations than genuine offender-years. Improving the detector helps, but it cannot abolish that arithmetic: unless specificity becomes extreme, false positives dominate; and as the threshold is tightened towards that point, sensitivity falls sharply.

The base rate. Deliberate fatal harm by healthcare staff is exceptionally rare, and the historical record is too sparse to establish a precise underlying rate. Since 1990, five UK nurses have been convicted of murdering two or more patients, for offences committed in the years shown: Beverley Allitt (1991), Colin Norris (2002), Benjamin Geen (2003–04), Victorino Chua (2011) and Lucy Letby (2015–16). Norris's, Geen's and Letby's convictions remain disputed; Norris's were upheld on a second appeal in 2025. Over the same years the register has held between about 620,000 and 810,000 nurses and midwives (and, before 2002, health visitors) (IES, NMC, NMC), some 25 million nurse-years in all. That is one convicted offender in about 5 million nurse-years. The 95% Poisson interval for this observed conviction rate is approximately one in 2 million to one in 15 million; it is not an interval for the true incidence of offending. It agrees with the "very roughly 1 in 2 million per nurse per year" that Gill, Fenton and Lagnado take from Forrest's 1995 catalogue. Each case ran for between two and thirteen months, so an offender counts as about one offender-year at most.

Two reference points. The national reference is that rate: one offender-year in 5 million nurse-years. The model's 8,800 nurses on 175 units are close to Health Education England's estimate of about 8,000 whole-time-equivalent neonatal nurses in England, once part-time working is allowed for. On that roster the national reference is about 0.1 offender-years per 10,000 unit-years. The neonatal reference takes the one neonatal case, Letby's, at face value. With roughly 175 units over the half-century since 1970, that is one case in about 9,000 unit-years, or about 1 per 10,000; counted only from 1990, it is about 1.6. That is ten times the national figure, but it rests on a single disputed case, and its interval spans two orders of magnitude. (Allitt's offences took place on a paediatric ward and are not counted.) Rates of 3, 10 and 30 per 10,000 are deliberately high-prevalence stress tests. Each unit's share of the risk is proportional to its nursing staff. Undetected offenders would push the true rate up and wrongful convictions push it down. The record measures neither, so these are scenarios, not estimates.

Detection. To measure sensitivity, I introduce an offender into each type of unit. The offender follows a full-time work pattern, causes an expected four or seven additional deaths over twelve months, and is present at every one of them. Each combination of rule,

unit type and effect is simulated 10,000 times. The simulation then asks two separate questions: how often does the monitoring rule flag the unit within that year, and, conditional on an alarm, how often does the subsequent rota analysis identify the offender, as the most-present nurse passing the own-exposure test? Table 3 gives the results.

Table 3. One-year detection sensitivity and offender identification, by monitoring rule and unit type

Rule	Unit detected: intensive care, +4 expected deaths	Local, +4	Special care, +4	Local, +7 (Letby-scale)	Rota review identifies offender: local, +4
A, annual 2-sigma	24%	40%	66%	71%	65%
B, annual 3-sigma	10%	17%	38%	44%	67%
C, rolling, monthly 2-sigma	40%	55%	73%	79%	34%
D, doubling heuristic	11%	67%	77%	86%	27%
E10, idealised, 1 in 10	25%	55%	83%	84%	46%
E50, idealised, 1 in 50	8%	23%	53%	60%	63%

Ties for maximum attendance are divided equally among tied nurses.

Combining detection and chance alarms across all 175 units gives the number of falsely flagged unit-years for each detected offender-year ("detection" below means the unit was flagged, not that the offender was identified; risk is in proportion to staff). Table 4 gives the results.

Table 4. Falsely flagged unit-years per detected offender-year, by monitoring rule and assumed base rate, with offender prevalence proportional to staff numbers

Rule	National reference, 0.1 in 10,000	Neonatal reference, 1 in 10,000	3 in 10,000	10 in 10,000	30 in 10,000
A	18,700	1,870	624	187	62
B	11,300	1,130	376	113	37
C	36,500	3,650	1,220	364	121
D	49,300	4,930	1,640	492	164
E10	22,900	2,290	764	229	76

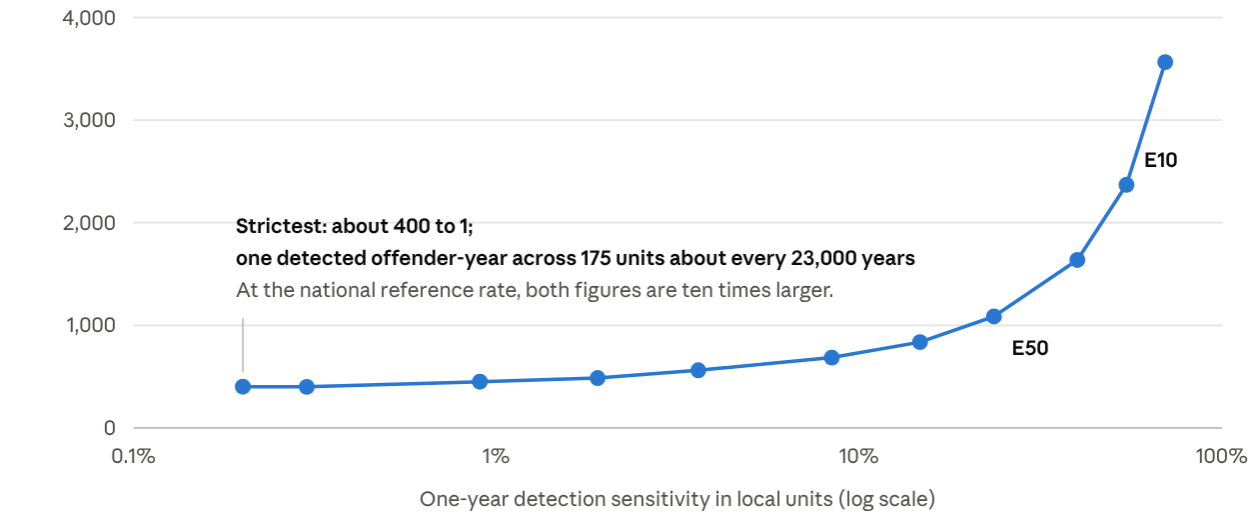
Rule	National reference, 0.1 in 10,000	Neonatal reference, 1 in 10,000	3 in 10,000	10 in 10,000	30 in 10,000
E50	10,900	1,090	362	109	36

None of the six rules achieves both high sensitivity to four extra deaths and a useful predictive value for deliberate harm. The strict rules (B and E50) keep false alarms down but detect a smaller proportion of offender-years. The sensitive rules detect a larger proportion, at the cost of about 1,900 to 4,900 falsely flagged unit-years per detection at the neonatal reference, and ten times that at the national one. At the neonatal reference, the 175 units produce one detected offender-year every 116 to 145 calendar years under C and D, or every 312 to 368 years under E50 and B. At the national reference, it is every 1,200 to 3,700 years.

The idealised chart shows the trade-off (Figure 2). A tighter threshold lowers false alarms and reduces the ratio, but detection falls almost as fast, and the ratio flattens. At the neonatal reference, it drops below 600 to 1 only when the chart detects fewer than 4% of local-unit offender-years within a year. Across the four strictest settings it moves only from about 490 to 400 to 1, and at the strictest the 175-unit system records one detected offender-year about every 23,000 years. At the national reference, both figures are ten times larger. Whether the ratio would keep falling at still stricter settings the simulation cannot say: calibration there rests on about 50 simulated crossings, so these tail results are approximate.

Figure 2. Stricter thresholds reduce the false-alarm ratio only at the cost of sharply lower detection sensitivity

Idealised risk-adjusted CUSUM, eleven settings, 175 units, neonatal reference rate of 1 in 10,000 unit-years



Simulation details: thirlwall_statistical_model.py, output block 9 · idealised CUSUM tuned to a doubling, calibrated on fixed null paths · +4 expected deaths · risk in proportion to staff · detection shown for local units, ratio for all 175 units

E10 and E50 are the two settings used in Tables 1 to 5. False alarms can always be driven lower by tightening the monitor. But even at the neonatal reference, no setting tested gets below about 400 false-alarm crossings per detection, and the settings that come close have very low one-year sensitivity.

Four further results are relevant. First, the rota review is weak in both directions. When there is an offender and the alarm is triggered, the review identifies the offender in only 27% to 67% of local-unit cases, and in 16% or fewer of intensive care cases, because the offender's deaths are diluted among the unit's natural ones. Second, giving every unit the same risk, instead of risk in proportion to staff, lowers every ratio by about a tenth to a third, because offenders in large units are harder to see. Third, giving the offender exactly four deaths rather than an expected four raises some ratios by up to about a fifth and lowers others by less than a tenth. In neither case do the conclusions change.

Fourth, detection is defined generously: any alarm in a year when an offender is present counts, whether or not the offender's deaths caused it. Running the same rules on the same kinds of unit with no offender shows how much of that "detection" is background. Under rule C, 18% of local units and 27% of intensive care units would be flagged in that year anyway, against 55% and 40% with an offender present. Under rule D in local units the figures are 33% and 67%: half of what counts as detection would have happened regardless. The idealised chart does better here, with 7% background against 55%. The predictive values above therefore, if anything, overstate the value of an alarm.

Putting the steps together. Let *q* denote the share of alarms followed by a rota search. The same share applies to every alarm, whether it arises by chance or from an offender, because the alarm alone gives no means of telling them apart. A manager who could select alarms on other evidence would change the odds below; finding that evidence is the job of the investigation this article recommends. Each unit keeps the same staff for the whole ten years, so the model can count distinct nurses as well as flagging episodes; the two barely differ, because repeat flags of the same nurse are rare. Table 5 counts the nurses each year whose record passes the own-exposure test when every alarm is searched, and scales that count by *q* for the other columns. A nurse flagged more than once has a better than *q* chance of being searched at least once, so this slightly understates the smaller columns, by at most 0.2 a year.

Table 5. Annual nurse flags under the own-exposure test, by rota-search probability (q)

Rule	q = 1	q = 0.5	q = 0.25	q = 0.1
A	6.2	3.1	1.6	0.6
B	1.8	0.9	0.4	0.2
C	16.4	8.2	4.1	1.6

Rule	q = 1	q = 0.5	q = 0.25	q = 0.1
D	16.2	8.1	4.0	1.6
E10	8.6	4.3	2.2	0.9
E50	1.8	0.9	0.5	0.2

The report does not set q at 1. Its data chapter treats a signal as a prompt for review, and Recommendation 9 is triggered by concerns about a person, not by a chart. Readers may substitute their own value of q. The one documented case is Chester itself, where the rota was examined because deaths had risen.

This is where a low threshold for suspicion becomes statistically consequential. With the offender on the rota, the model can also say how likely it is that a flagged nurse is in fact the offender. At the reference rates, it is between about 0.001% and 0.1%, depending on the rule. Even at the neonatal reference, only one flagged nurse in 1,000 to 8,000 is the offender. In the highest-prevalence case tested, the strictest risk-adjusted chart at 30 in 10,000, it reaches 3.0%, or 5.0% if every unit carries the same risk. Recommendation 9 treats any good-faith concern as enough to act on, whatever the manager believes, and the report calls the action neutral.

Costing the consequences

The principal cost falls on staffing, already the most constrained resource in neonatal care. Camera schemes operate at substantial scale for a very low expected detection yield.

Removals. Suppose each nurse who passes the rota test is removed from clinical duties while an investigation proceeds. Dr Martin Pitman's own investigation was promised at six weeks and took eighteen months. At that pace, rule C with every alarm followed up ($q = 1$) would keep about 25 nurses off neonatal wards at any one time; at $q = 0.25$, about 6. Those figures cover data-driven alarms only. Concerns raised by colleagues, which the protocol also covers, cannot be modelled and may well exceed them.

These losses fall on units already under staffing pressure. In 2022 only 71.1% of nurse shifts in England, Wales and Scotland were staffed to recommended levels, and the figure had fallen for two years running (Infant journal, on the NNAP report). Staffing matters to survival: a 2007 study found that bringing specialist nurse ratios up to one-to-one was associated with a 48% fall in risk-adjusted mortality among the smallest babies (Bliss evidence to Parliament). A safeguard that removes nurses on chance alarms may

therefore reduce the staffing on which patient safety depends. Potential effects on recruitment and retention, if nurses come to believe that a statistically unremarkable cluster could end a career, are plausible but are not quantified here.

Cot cameras. Recommendation 1 calls for in-cot cameras with livestreaming video on every cot and incubator, so that parents can observe their baby remotely (Thirlwall Inquiry). The North West has 469 cots across 21 units, about 22 a unit (NHS England North West), which implies roughly 3,500 to 4,000 cots in England and Wales. At the reference rates, 175 units over a decade imply an expected 0.02 to 0.2 offender-years, however many cots carry a camera. The 37,000 cot-years of camera coverage measure the scheme's scale, not its likelihood of detection. Any case for the scheme must therefore rest substantially on benefits other than expected detection yield, such as reassurance and deterrence.

Insulin CCTV. Recommendation 2 calls for cameras on insulin storage, with recordings kept for at least 28 days, until digital access controls are in place (Thirlwall Inquiry). Such recordings help only if suspicion arises within that window and the insulin came from that store. A central dispute in the Letby insulin cases concerns something else: an immunoassay result that was never confirmed by a more specific test. Routine mass-spectrometry confirmation of unexpected insulin results would protect against wrongful accusation as well as against poisoning.

Implications for practice

Monitoring is worth having if an alarm leads to a review of care, not a search for an individual. The essential safeguard is a statistical one, placed between the signal and the suspicion.

1. **Set the rule nationally, with a stated false-alarm rate.** The chart and its threshold should be specified nationally, with appropriate statistical expertise and adjustment for each unit's case mix, rather than determined independently by each trust. The expected number of chance alarms can then be estimated in advance.
2. **Treat an alarm as a question about care.** The first response should examine infections, staffing, case mix and equipment, which are among the alternative explanations for any cluster. Paul Aylin, who built the post-Shipman monitoring system, argued that monitoring should start audit and learning, not recrimination (Digital Health).
3. **Require independent expert review before a statistical alarm is translated into person-focused suspicion.** The Royal Statistical Society recommends that such investigations be supervised by panels independent of both the suspect and the employer (RSS). It is the kind of panel the National Crime Agency is reported to have advised in the Letby investigation, and which was never convened.

4. **Blind the reviewers.** Classify each death as explained or unexplained before anyone sees the rota. Fix the list of events and the time window first.
5. **Count the whole rota.** If staff presence is examined at all, compare every nurse's record and allow for the fact that someone always comes top.
6. **Separate protection from accusation.** Where a concern is serious, a change of duties agreed with the nurse, without a finding, protects patients without implying any finding of fault. It is closer to the "neutral act" the report describes than suspension is.

Synthesising the signals

Taken together, the three stages show how a mortality signal can acquire more apparent evidential weight than the underlying statistics justify. Chance alone produces several to dozens of alarms each year. If the deaths associated with those alarms are then matched retrospectively to staff attendance, the most-present nurse appears statistically significant in about half of cases under the own-exposure test, even though staff presence and mortality are independent in the simulation. Risk adjustment improves the monitoring stage but does not resolve the underlying base-rate problem. At the two reference rates, falsely flagged unit-years outnumber detected offender-years by at least a thousand to one under every rule in Table 4. More stringent thresholds reduce that ratio, but only at the cost of sharply lower sensitivity. The rota analysis is also weak in the opposite direction: when an offender is present, it often fails to identify the offender.

The risk therefore does not arise from any one recommendation in isolation. Mortality monitoring, action on credible concerns and independent expert review are each defensible. The statistical vulnerability appears in the transition between them: the point at which an abnormal outcome pattern becomes a concern about a particular person. The recommendations do not specify a statistical safeguard for that transition. How often trusts make it is a behavioural question and is therefore left as an explicit parameter in the model; the consequences once they do can be analysed directly.

The appropriate response to a mortality signal is therefore an independent investigation of care before any person-focused inference is made. Staffing, workload, infection, equipment, clinical practice and case mix should be examined first; statistical thresholds should be specified in advance; and any subsequent analysis of staff attendance should be conducted against the complete rota rather than around a previously selected individual. The purpose of monitoring is to identify where further investigation is warranted. Attribution will always require additional evidence.

A mortality signal can help identify an abnormal pattern. It cannot, on its own, determine the cause or attribute it to an individual.

Assumptions and limits

The numbers are illustrative, not forecasts. They show orders of magnitude; the reported sensitivity checks show which conclusions persist as assumptions change.

- **Unit mix, death rates and staffing are modelled.** The split of 45 intensive care, 85 local and 45 special care units, their average deaths and their staff numbers (100, 40, 20) are my assumptions, informed by published national totals. Real units may differ and produce different alarm rates.
- **The monitoring rule is unknown.** The report leaves it open. Rules A–D are deliberately simple. Rule E is an idealised CUSUM benchmark with knowledge of the true monthly expectation, an advantage a real monitor would not have.
- **Baselines are recent history.** Rules A–D use the three years before each window; E uses the true monthly expectation. Annual rates vary, but long-run means are fixed.
- **Rotas are simplified.** Each nurse's presence at each death is independent of everyone else's. In reality deaths cluster at night and on busy shifts, and some nurses work more of those. That can make chance coincidences more likely, not less.
- **Behaviour is a parameter, not a forecast.** How often an alarm leads to a search for a person (q) is left open and shown at four values. It is assumed to be the same for every alarm.
- **The base rate is anchored, not estimated.** The national reference rests on five UK convictions since 1990 and published register totals; the neonatal reference rests on one disputed case. Undetected offenders would raise the true rate, wrongful convictions lower it. Stress tests up to 30 per 10,000 are shown, and the conclusion holds across all of them. Risk is spread in proportion to staff; equal risk per unit gives somewhat lower ratios and the same conclusion. The offender is modelled as a full-time nurse present at all the extra deaths; nothing else sets this nurse apart.
- **Paediatric wards are left out.** The report's monitoring covers deaths of children as well as babies, which would add further alarms.
- **Detection is defined generously.** Any alarm in a year when an offender is present counts as detecting the offender's unit, whether or not the offender's deaths caused it. Step three shows how much of it is background.
- **Units of the false-to-true ratio.** Table 4 compares falsely flagged unit-years with detected offender-years; the threshold sweep in Figure 2 counts chart crossings instead. These are distinct measures and should be compared with that difference in mind.
- **Chance a flagged nurse is the offender.** The offender side counts the first alarm in the offender's year, while the background side counts every flagged episode. This asymmetry affects interpretation of the resulting probability, though a check during

review, using matched event definitions, found its numerical effect negligible at these base rates.

- **Ties in rota attendance.** Where several nurses share the maximum attendance count, identification probabilities are apportioned equally among the tied nurses. This is the expected result of a random tie-break and avoids dependence on roster ordering.

Testing the assumptions

The tables in this article are a snapshot: one run of the model under one set of assumptions. Each assumption is a choice, and a reader may reasonably prefer another. The interactive explorer is built for that. It follows the argument in five steps, one for each table: how much noise each rule makes; how that noise becomes a named nurse; what each rule does when an offender is present; the ratio that combines the two; and the human cost.

In its default view, the explorer reproduces every table exactly, from the same reference run. Readers can change the base rate, the share of alarms searched, the allocation of risk and the number of units, all of which the published results determine exactly. They can also change the world itself: deaths a year, roster size, the alarm line, the significance level and the offender's effect. Those settings are simulated afresh in the browser by a version of the model checked against the original, and each result is shown with its uncertainty beside the paper's value. The explorer cannot say what the true base rate is or how managers will behave. It shows how much each answer depends on them.

The underlying simulation is documented in the source code and technical notes, with the full specification in MODEL.md. It is one explicit model and interpretation, open to comparison with others. Its versions are listed in the README's Version history, and the changes made to this article in each round of review are recorded in a separate technical revision history.

Terms used

A unit-year is one neonatal unit monitored for a year, a nurse-year one nurse employed for a year, and an offender-year one year in which an offender is at work on a unit. A detected offender-year is one in which an alarm is triggered on the offender's unit; it does not mean the offender was identified.

Disclaimer

This analysis is offered in good faith, on a best-efforts basis, as one interpretation of public material. It is not legal, medical or other professional advice, it makes no finding about any individual or case, and it may contain errors or omissions. The author accepts no liability for any decision made in reliance on it. Corrections are welcome.

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